Unit 11 Alcohols and Ethers

Key answer

1. $2\ R - OH + 2Na \rightarrow 2\ RONa + H_2\uparrow$ 2 moles of alcohol gives 1 mole of H_2 which occupies 22.4L at 273K and 1 atm

$$\therefore \text{ number of moles of alcohol} = \frac{2 \text{ moles of R - OH}}{22.4\,\text{L of } H_2} \times 560 \text{ mL}$$

$$= 0.05 \text{ moles}$$

$$\therefore \text{ no. of moles} = \frac{\text{mass}}{\text{molar mass}}$$

$$\Rightarrow \text{ molar mass} = \frac{3.7}{0.05} = 74 \text{ g mol}^{-1}$$

General formula for R - OH $C_n\ H_{2n+1}$ - OH

$$\therefore n(12) + (2n+1)(1) + 16+1=74$$

$$14n = 74 - 18$$

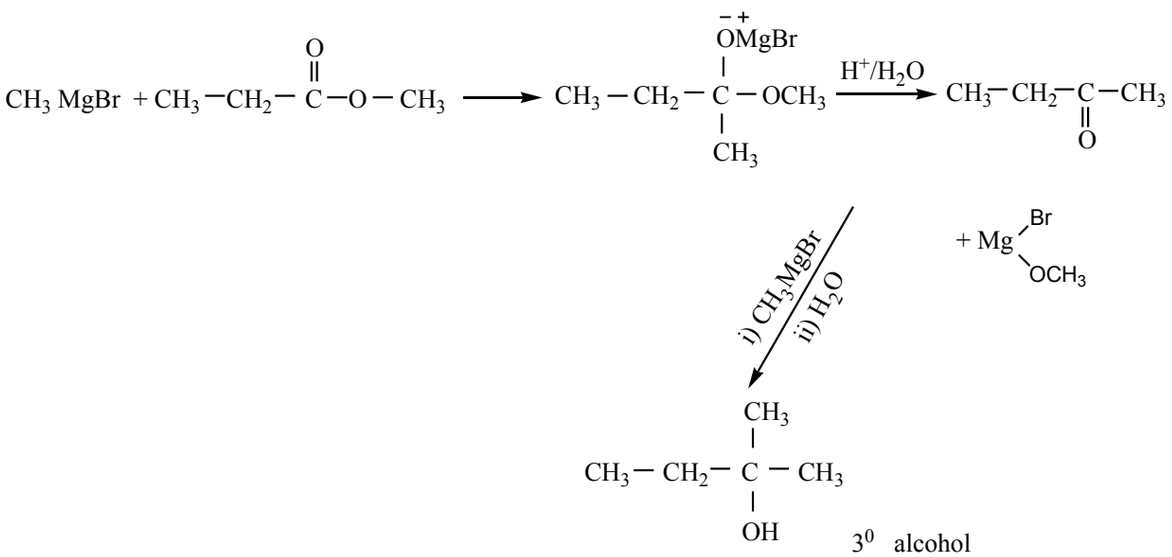
$$14n = 56$$

$$\therefore n = \frac{56}{14} = 4$$

The 2° alcohol which contains 4 carbon is $CH_3CH(OH)CH_2CH_3$

Option (a)

2.



Option (c)

3. Hydro boration – Anti markownikoft product i.e., $CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - OH$

Option (a)

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